

23. a. With neat sketch, explain the construction and working of stepper motor. 8 2 3 1

(OR)

b. Explain in detail, the construction and working of PMSM. 8 2 3 1

24. a. Explain the construction and working of attraction type and repulsion type moving iron instruments. 8 2 4 1

(OR)

b. Write short notes on
(i) Photodiodes
(ii) Photovoltaic cells 8 2 4 1

25. a.i. Draw the single line diagram of typical generation, transmission and distribution of AC power system. 3 2 5 1

ii. Briefly discuss on various substation equipments. 5 2 5 1

(OR)

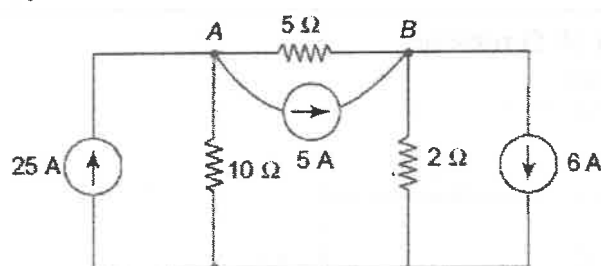
b. What is the need for earthing? Explain pipe earthing with neat diagram. 8 2 5 1

PART – C (1 × 15 = 15 Marks)

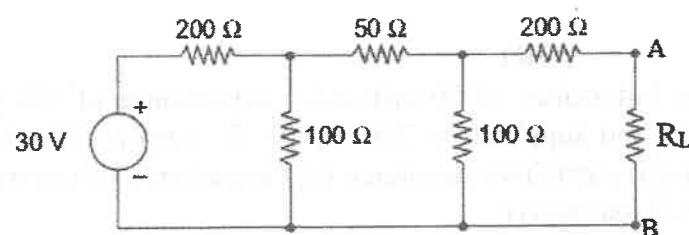
Answer ANY ONE Questions

Marks BL CO PO

26. Using nodal analysis, find the node voltage V_A and V_B . Also find the power observed by each resistor. 15 4 1 2



27. Using maximum power transfer theorem, find the maximum power consumed by load R_L . 15 4 1 2



Reg. No.

B.Tech. / M.Tech (Integrated) DEGREE EXAMINATION, JANUARY 2023
First Semester

21EES101T – ELECTRICAL AND ELECTRONICS ENGINEERING
(For the candidates admitted from the academic year 2022-2023)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B and Part-C** should be answered in answer booklet.

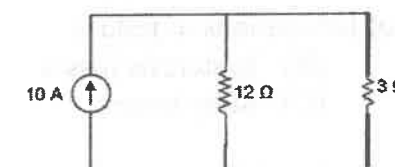
Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

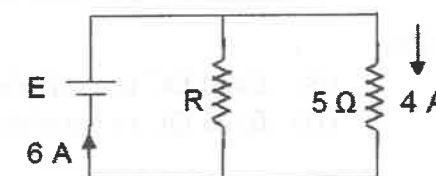
Answer ALL Questions

1. In the circuit shown below, find the current flowing through 12 Ω resistor.



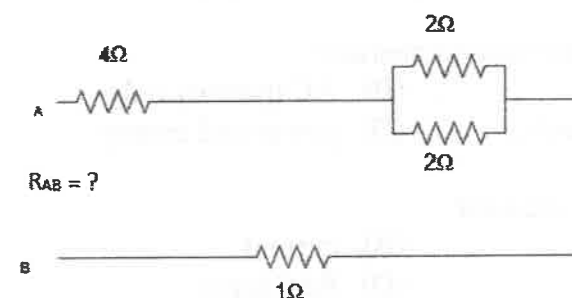
- (A) 8 A (B) 5 A
(C) 2 A (D) 4 A

2. What is the value of R in the circuit shown



- (A) 2.5 Ω (B) 5 Ω
(C) 7.5 Ω (D) 10 Ω

3. Find the resistance across the terminals A and B



- (A) 6 Ω (B) 9 Ω
(C) 0.833 Ω (D) 4.5 Ω

4. For a three phase delta connected system, the line voltage is 1100 V and line current is 500 A, Calculate the value of phase voltage and phase current.

- (A) 1100 V, 500 A (B) 1100 V, 288.67 A
(C) 635.08 V, 500 A (D) 635.08 V, 288.67 A

5. Which of the following is not a terminal of BJT? 1 1 2 1
 (A) Emitter (B) Base
 (C) Collector (D) Gate
6. The Algebraic function for Ex-NOR Gate is 1 2 2 1
 (A) $A'B+AB'$ (B) $AB+A'B'$
 (C) $AB+A'B$ (D) $AB+AB'$
7. The minimal SOP for the function $Y(A,B,C) = \sum m(0,1,2,3,4,5,6,7)$ is 1 3 2 1
 (A) 0 (B) 1
 (C) A (D) A'
8. FPGA stands for 1 1 2 1
 (A) Flexible Programmable Gate Array (B) Field Programmable Gate Acceleration
 (C) Flexible Programmable Gate Acceleration (D) Field Programmable Gate Array
9. The transformer core is made up of laminations to reduce 1 1 3 1
 (A) iron losses (B) hysteresis losses
 (C) eddy current losses (D) Stray losses
10. In a three phase induction motor, the three phase winding R,Y,B are connected in 1 2 3 1
 (A) series (B) parallel
 (C) series and parallel (D) star
11. A chopper is a device that converts 1 1 3 1
 (A) fixed AC to variable AC (B) fixed DC to variable AC
 (C) fixed AC to variable DC (D) fixed DC to variable DC
12. A step down transformer has a primary voltage and current of 1000 V, 20 A. Its secondary voltage and current are 1 3 3 1
 (A) 100 V, 200 A (B) 10000 V, 2 A
 (C) 10000 V, 20 A (D) 100 V, 20 A
13. Moving coil instruments are used to measure 1 1 4 1
 (A) DC quantity only (B) AC quantity only
 (C) both AC and DC quantity (D) power and energy
14. Digital multimeter cannot measure 1 1 4 1
 (A) voltage (B) current
 (C) power (D) Resistance
15. A LVDT is used to measure 1 2 4 1
 (A) frequency (B) displacement
 (C) temperature (D) speed

16. The symbol represents _____ device



- (A) LDR (B) LED
 (C) LCD (D) Photo resist transducer

17. What is the highest AC transmission line voltage in India?

- (A) 400 kV (B) 765 kV
 (C) 1100 kV (D) 1265 kV

18. Which of the following is not a renewable energy resource?

- (A) Solar power (B) Wind power
 (C) Thermal power (D) Tidal power

19. What is the most common fuel used in fuel cells?

- (A) Water (B) Hydrogen
 (C) Nitrogen (D) Sulphur

20. Which types of battery is highly preferred for EVs?

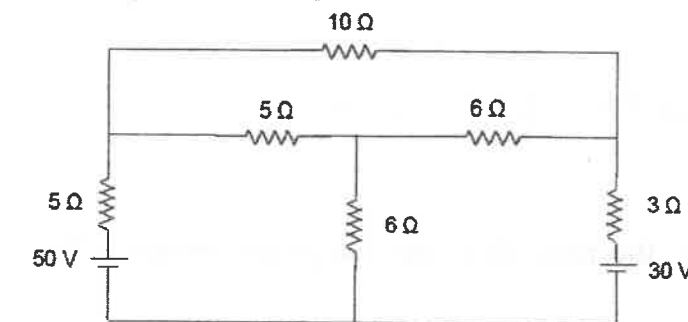
- (A) Lead acid battery (B) Nickel cadmium battery
 (C) Nickel cadmium battery (D) Lithium ion battery

PART – B (5 × 8 = 40 Marks)

Answer ALL Questions

Marks BL CO PO

21. a. Find the current through 10 Ω resistor.



(OR)

- b. A resistor of 10 Ω, an inductance of 50 mH and a capacitance of 150 μF are connected in series and supplied by 200 V, 50 Hz supply. Calculate (i) inductive reactance (ii) capacitive reactance (iii) impedance (iv) current (v) power factor and (vi) real power.

22. a. Explain the working of SCR. Also draw its VI characteristics.

(OR)

- b. Find the minimal Sum of Products for the Boolean expression, $Y(A,B,C,D) = \sum m(1,2,3,4,9,10,11,12,15)$ and implement it using Gates.